

PV207 course manual

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1 How to use this document

1. **This document should provide ALL the information you need to know about the course.**
2. **Before you ask any question, make sure the information is not here** (full text search, look in FAQ)
3. If you are sure the information is missing
 - a. **use comment (in this document)** to ask a question,
 - b. **You have to be signed-in with your google account**, anonymous questions may be deleted.
 - c. **it will be answered** in the comment and you get a notification.
4. If there is a **major update** of the document, you will be **notified by email**. For minor cosmetic changes, the notifications are not sent. Before any course milestone check this document for updates.

2 Course goals (Why to study PV207 BPM)

After this course you should

- Understand what BPM is
- Understand where it can be helpful and where NOT
- Understand how a BPM-based project should be organized
 - Teamwork and roles
 - Organization
 - Conditions and environment
 - Common problems
- Know some basics of Business analysis
- Get a basic overview of recent BPM technologies
- Team project goals
 - Improve teamwork skills
 - Understand different roles in BPM
 - Learn about whole process life-cycle
 - Learn how to bridge the gap between analysis and implementation
 - Exercise domain analysis
 - Exercise precession in analysis documents
 - Hands-on BPM related technologies

3 Course rules

1. **Interaction is appreciated**

Interaction and involvement in the course contributing to the course quality is being recognized and being taken in account during evaluation.

2. Fairness and equal conditions

We try to assure fairness by setting the same conditions for all students, therefore any non-standard requests that violate the equal conditions for everybody are not welcomed

3. No-go for Cheating

ANY form of **cheating**, **plagiarism** or **misbehavior** is carefully **watched** and **prosecuted by** requesting **termination** of your studies.

4. Feedback is collected

Course is partially student-driven, **Anonymous feedback** is collected at lectures

- a. Be honest, be constructive
- b. If you don't like something, propose how to do it better
- c. Express wishes, but consider if they are realistic

5. AI use Guidelines

- a. We generally follow [University AI guidelines](#) but we have different rules for:
 - i. **Implementation and coding**: You can use AI for code generation as much as you want. Just clearly state which code was generated.
 - ii. **BPMN process modeling**: You can use AI if you want, but you have to attach **detailed description text/code how AI was used** to generate the deliverable.
 1. which model, what service,
 2. step-by-step procedure of AI use.
 3. Full content of all prompts
 - iii. **Text outputs and Analytical documents**: DO NOT use AI. As for now humans seem able to elaborate Analytical documents in better quality than AI. We believe you can do better as well.
- b. **Sharing** prompts and AI instructions used for output generation among students is considered **plagiarism** the same way as sharing the outputs
- c. Presenting AI generated outputs as human work is considered **cheating**
- d. You are still **obliged to understand in detail everything what AI did** and be able to **do the same without AI** (this can be verified during Examination and Homeworks checks)

b4 Course schedule

Calendar is the **MASTER source** for course schedule (NOT IS MUNI). Only **exception is the exam schedule, where IS MU is the MASTER source (faculty policy)**. Always synchronize it and check regularly.

- Google calendar - Agenda WebView
 - <https://www.google.com/calendar/embed?src=aq8rri3foddsn7k7ho586tl7ms%40group.calendar.google.com&ctz=Europe/Prague&mode=Agenda>
- Add the calendar to your google account or/and Phone: Click Other calendars > Add by URL, enter following URL:
 - <https://www.google.com/calendar/ical/aq8rri3foddsn7k7ho586tl7ms%40group.calendar.google.com/private-446df1f86ae2a32ef22f5b739634575b/basic.ics>

5 Course components

5.1 Lectures

- Presentations with slides (will be provided after each lecture)
- Practical examples (audience activity needed)
- Given in English
- External guest lectures
- Discussions more than welcomed

5.2 Seminars

- **Students are being assigned to the teams and work in teams**
- **You can have 2 absences during the semester (2 seminar sessions). Not more.** You are responsible for gathering the information you missed during your absence. Agree on the absence with your **team**
- **It is your responsibility to find and sign a presence-arch on every seminar!**
 - **No signature, no presence evidence.** (Exception is only when presence is not recorded, this have to be clearly stated by the lecturer, if you are not sure, ask during the session)
- Seminars include
 - Hands-on sessions with PC
 - Modeling, tools & coding,
 - Team activities
 - Homework & Team project consultations
- Student **active involvement in seminar session is crucial**
- Used Technologies: Java, Camunda, Signavio academic platform, Excel, win/mac/linux

5.3 Homeworks

- Assignments are published during the process modeling seminar (see calendar for exact date)

- Completed homework should be
 - uploaded to Homework Vaults in IS MU (**PDF + Camunda readable .bpmn/xml**)
- Results are published in one of the following seminars (see calendar for exact date).
- **Homeworks are your Individual work, not teamwork**
- **Plagiarism is harshly punished.** share knowledge, learn from each other, but:
 - do not copy from others,
 - do not elaborate homework for other people
- Homework 1:
 - Install and run Camunda, model and run basic process
 - few days time for elaboration
 - Results are published on seminar + discussion
- Homework 2:
 - basic modeling
 - few days time for elaboration
 - Results are published on seminar + discussion
- Homework 3:
 - complex modeling
 - cca 2 weeks time for elaboration
 - 1 consultation on seminar prior to submission
- SEE THE **CALENDAR FOR DEADLINES**
- **DEADLINES ARE SHARP**

5.4 Project

5.4.1 Recommended schedule

Starts with team-Building at the beginning of the semester.

Ends with project presentation during the Exam period.

- Phase 1: Domain analysis
 - Recommended: ~9.3~30.4.
- Phase 2: Process analysis
 - Recommended: ~30.3~11.5.
- Phase 4: Presentation
 - Preparation: Recommended: ~1.5~18.5.
- Examination
 - ~20.5~22.6.

See calendar for exact deadlines

5.4.2 Teamwork

Students are assigned to the team based on Team Questionnaire filled at the beginning of the semester.

- Project teams have 4 member teams, with exception of the last one or two (div 4)
- **No later rearrangements of teams**
- Success in the course ~ **success of the whole team** ~ TEAM RESPONSIBILITY
- **Work on all activities together**, however, have a **main responsible person** for each activity:

Project manager

- Management - organizing the teamwork (1 person)
- Communication with lecturers and tutors
- Coaching

Personal skills:

- Soft-skills, authority, responsibility

Business analyst

- "Expert" in domain you are going to analyze
- Accuracy in writing analytical documents
- Understanding of basics of strategic planning and
- business analysis

Personal skills:

- Accuracy, responsibility, domain knowledge

Process analyst

- Good knowledge of process modeling and BPMN
- Good understanding of the domain

Personal skills:

- Accurate, communicative, solution oriented

BPM/Integration developer

- Good understanding of process modeling
- Technical skills
- Chosen BPMS
- Web services programming

Personal skills:

- Patience, accuracy, technically skilled

5.4.3 Project activities (types of work)

5.4.3.1 Business analysis

- Tasks
 - Learn about your domain and context
 - Collect real-world information about domain
 - Define Strategy , mission and vision of your organization
 - Define goals, objectives and per indicators
 - Define structure of your organization
- Roles involved:
 - Mainly work of business analyst
 - Discussed with all team members
 - Agreement of whole team
- Deliverables
 - Up to 5 lines describing context of your organization
 - cca ¼ page describing strategy=[vision,mission etc.]
 - Simple clear and expressive
 - Your goals should be based on your vision, but do not repeat the vision in goal definition
 - 2+ well defined and described goals
 - 10+ well described objectives linked to goals
 - Description of KRI/RI/KPI/PI linked to G&O
 - Description of organization structure (text or tree)
 - Roles and responsibilities
 - Departments and responsibilities
 - **USE THIS TEMPLATE** , you can do minor modification of the template if needed. Keep in mind, that the document has to be understandable for a person who had never seen the template.

5.4.3.2 Process analysis

- Tasks
 - Identify important processes in your organization
 - Link processes to your goals and objectives
 - Define measurement of your indicators on processes
 - Describe your processes in detail
- Roles involved
 - Process analyst, Business analyst
 - Validate with all team members
- Deliverables
 - 8+ identified modeled/documented processes
 - Linked to G&O

- Linked to indicators
- Short text description for every process
- Short description of every data object used in the in process
- Each process has 4+ activities
- Valid BPMN 2.0 Level 2 for every process
- Implemented processes are matching to processes in process analysis

5.4.3.3 Implementation

- Tasks
 - Build an interface catalog
 - List at least 5 different system connections and their interfaces
 - Choose transmission protocols, data formats, data objects and interaction types / behaviour
 - Call the systems in your business processes - they don't need to be implemented (mock using a generic service task / script task)
 - Implement processes
 - Implement some service stubs (service tasks)
 - Test your solution
 - User side testing
 - Do backup for presentation (e.g. another laptop)
- Roles involved
 - Process developer, Process analyst
 - Validate with all team members
- Deliverables
 - Interface catalog in the analysis project document

Implementation of 4 executable processes containing (following numbers are sum for all processes in the project, not per process):

 - Camunda SaaS (online version, use 30-day free account)
 - We encourage you to create a project in Camunda SaaS and then create a Process Application within that project, it will make your project organization much easier
 - Integration of 3 external services
 - Can be of various types
 - REST (Public, mocked, custom. Public API is preferred, e.g. weather API or own REST application (Spring Boot, Quarkus, Node.js...))
 - DB services
 - Rules
 - IFTTT...
 - Every (meaningful) REST endpoint counts as a separate service
 - On top of these 3 services:

- Implement at least 1 integration service based on Camel (not counting towards 3 external services)
 - Make a working service call to the Camel integration service from a business process
- Email interaction (not counting towards 3 external services)
- At least 1 Camunda Job Worker which would integrate meaningful external service
 - 10+ human tasks (4 full featured forms) in total
 - 15–30 activities (based on the nature of the processes) in total
 - 4+ intermediate events (signal, timer, error, link) - make sure some of those are included in the process you'll be demoing
 - At least one sub process (either embedded or reusable)
 - [BONUS] DMN integration will be awarded with bonus points
- Bizagi - Requirements will be provided upon request
- IBM BPM - Requirements will be provided upon request
-

For all the processes prepare testing data!

- Make sure your 30-day evaluation period of Camunda doesn't expire before your presentation. After 30 days your project will still be accessible but you won't be able to execute processes. Process execution is a mandatory part of your presentation. In case your evaluation period expired, create a new account.

5.4.3.4 Project submission

- **48h prior** a presentation into the “**project**” folder in IS MU a team member submits:
 1. **Presentation as .pdf** (projectName_presentation.pdf) Use provided template
 2. **.zip with all process models as a .bpmn/xml files** readable in Camunda Modeler
 3. **Analysis document pdf** (projectName_analysis.pdf)
 4. **zip with documented deployable implementation, and short howto** (projectName_implementation.zip)
 - a. In your Camunda SaaS project, click on the 3 dots to the right of your Process Application and click Download. Put this downloaded zip file into the implementation zip file.
 - b. External REST service(s) if you implemented them yourself (Spring Boot, node.js app...) with a readme file containing steps on how to run them
 - c. Either invite email bpmtestfi@seznam.cz as a collaborator to your Camunda SaaS project (Open your project -> click on Collaborators tab on the far right -> invite the email as a Project Admin) OR share

credentials to your Camunda account within the implementation zip file so we can access the project online.

- v, export as PDF

5.4.3.5 Presentation

- **F2F in the scheduled room.**
- **Be on time all team members, no exceptions beside serious illness, it's an exam ;)**
- Presentation and live demo will be streamed from **your laptop**
 - **make sure you can connect with HDMI, it is your responsibility to test your laptop, rooms are publicly accessible any time...**
 - **Make sure faculty WiFi works, (have a backup?)**
- **BRING 1 PRINTED** copy of your analysis document (simply binded papers, no plastic)
 - **PROCESS DIAGRAMS** are obviously part of this document, **HAVE TO BE READABLE ON THE PAPER** (hardly readable process will be decreased by $\frac{1}{3}$ of score)
- 15 minutes presentation (whole team! Timing! Prepare! ;) with the structure:
 - Project & members introduction (roles) - 1 min
 - Each member will explain what he did - 5mins (1 min / member)
 - Demo implementation highlights (the best process?) - 5 min
 - Questions, discussion, 1:1 rest of the demo with Marian - 15 min
- Presentation **can be in English**. For full CZ/SK teams English is welcomed, but not required
- **USE [THIS TEMPLATE](#), do only cosmetic modifications**

6 Evaluation and examination

6.1 Examination

Examination is done during the faculty exam period.

There will be three examination days

Exam consist of:

- Project presentation (WHOLE TEAM! 15 mins)
 - **Team Leader books the slot in IS** (One slot per team ~ 4 members)
 - Read details in [Project / presentation section](#)

6.2 Examination slots and project registration

To be able to present your project, you have to be registered at 2 places:

6.2.1 Examination slots (IS MUNI)

- Examination slots in IS, opens around may (see IS).
 - ONLY TEAM LEAD registers the presentation slot for whole team,

6.2.2 Project registration (gForms)

(Do less than 30 days prior the calendar deadline)

In addition to IS registration, Your Team Lead is responsible for filling out the following registration form <https://forms.gle/UEGFTvwzDDcZPtZWA> for the whole team and enter basic information about your project and team members.

Save the modification link which appears after submission of the form for later modifications. Before the submission, agree on a domain of your project and eventually update your project information later. DEADLINE for registration of your project in Google Forms is in course calendar. Further modifications are allowed any time.

6.3 Evaluation components

Your score is expressed in %. You can score up to 100 % in this course. The 100 % is a sum of following components:

1. Homework 1: Max **5 %**
2. Homework 2: Max **10 %**
3. Homework 3: Max **15 %**
4. Team project **70 %**

! Scoring less than 6/10 of any single components above means immediate FAIL (F)

Example: You get only half of the points from Homework 2: that means 5/10 of the score for the “Homework 2” component, you get F and you do not continue further in the course (such way you may get a grade prior to the examination period ;) But no worries, Homework 2 is quite easy if you study a bit.

6.4 Grades

- A 100–90 %
- B 89–86 %
- C 85–81 %
- D 80–76 %
- E 75–71 %

- F < 71 % (FAIL)

8 Information sources & books

- **Lecture slides** (not completely sufficient for the exam, lecture attendance is highly recommended)
- **Books @ Library FI & ESF MU**
- **Books @ Amazon**
- Papers & publications
- Extra books (beyond the course border)
 - BPMN method and style, *Bruce Silver*, 20099780982368107
 - Business Process Management: Practical Guidelines to Successful Implementations
 - Business Process Management: A Rigorous Approach
 - Business Process Management: Concepts, Languages, Architectures
 - Essential Business Process Modeling
 - Business process management: the third wave, *Smith, H. andingar, P.*
 - The Art of Business Process, *Schedlbauer, M.*
 - Podnikové informační systémy, *Josef Basl, Roman Blažíček*, 2007, 978-80-247-2279-5
 - Podnikové procesy: procesní řízení a modelování, *Václav Řepa*, 2007, 978-80-247-2252-8
 - Zavádění a rozvoj procesního řízení ve firmě, *Filip Šmída*, 2007, 978-80-247-1679-4

9 Contacts

Please look for a right person for your problem

Jiří Kolář

- Duties: Organization, general BPM lectures
- Primary contact for: **Organization, evaluation, General project Q&A**

Lubomir Hruban

- Duties: Process modeling, seminar-sessions
- Primary contact for: **BPMN Q&A, homeworks**

Lukáš Smiga

- Duties: Technical lectures, process modeling, seminar-sessions
- Primary contact for: **BizAgi platform, BizAgi based - projects**

Marián Macik

- Duties: Technical lectures, process modeling, seminar-sessions
- Primary contact for: **Camunda platform, Camunda based - projects**

External speakers

- IBM SWG CZ
- Red Hat Czech (Camunda)
- Lubomír Dočkal - Process modeling specialist

10 General FAQ

In which language will be course given?

Lectures are going to be given in English and all materials are in English as well. There is one English seminar group and one Czech. Your outputs can be any reasonable mix of Czech/Slovak/English

Are the lectures going to be recorded?

Only the first one, later lectures not. This is on-site course, not online course. If you want to change this, propose to faculty to start with online study programmes.

10.1 Where I can find more information about BPM?

There is a plenty of sources. For the beginning, you can look at More information/Information sources in this document. Another option is to ask one of the lecturers.

10.2 What can I do if I have a question that is not answered here?

Insert a comment in the document as it is described in the first section.

10.3 What should I do if one of team members stops responding?

A team leader will try to manage the communication problem himself at first. If the problem persists for more than a week, he informs his lecturer. During these days he has to assure not delaying the work on project. It is important to inform about this situation at final presentation.

10.4 What if there is a disagreement in your team?

A team leader is responsible for solving this problem (it doesn't mean that a team leader has to solve it on his own). Knowledge from other courses or from real situations can be used. You can ask for an advice any of the lecturers as well.

Why do we have to handle printed Homeworks?

We would also love to be green, but our rule is: Properly modeled process is a process which you can print on a paper, bring it to a meeting and open discussion about it. So HW is also an exercise on this. Believe or not, every year we have many people who fail to bring readable processes to their project presentation...

Would it be possible to share the slides before the lecture? I'd like to use them to take notes during the class.

For some lectures this will be possible and you can find them in IS, but we can not guarantee every week due to side updates.

I wanted to ask about the project template: KPI/KRI/PI/RI - does it mean that we only choose one of them, or do we need to have all four for each objective?

jBPM Implementation FAQ

How can I make a call to an external REST service?

1. Install Rest work item handler to your project using Settings->Custom Tasks Menu.
 - 1.1. Click Install.
 - 1.2. Provide User Name and Password.
 - 1.3. Refresh perspective by going to Assets and back to Settings.
2. Go to Deployments->Work Item Handlers and modify the Work Item Handler by adding classLoader parameter to it.
 - 2.1. It should look like this:
`new org.jbpm.process.workitem.rest.RESTWorkItemHandler("username", "password", classLoader)`
 - 2.2. classLoader parameter will allow the Work Item Handler to load the class which will be the object representation of JSON/XML response returned from the called REST service. This class will be created in the next step as a part of your project.
3. In your project, add an asset of type Data Object, which will be the object representation of your JSON/XML response.

4. Your object should be simple enough to be automatically transferable from JSON/XML to Java Object and back. For the PersonResponse class below the corresponding JSON structure would be:

```
{
  "name" : "John Doe",
  "age" : 33
}
```

Or XML

```
<?xml version="1.0" encoding="UTF-8"?>
<com.myspace.test.TestDao>
  <name>John Doe</name>
  <age>33</age>
</com.myspace.test.TestDao>
```

5. Finally, you have to make your Data Object available to jBPM remote services.
- 5.1. Click on the name of your class.
 - 5.2. Select Drools & jBPM Menu on the right.
 - 5.3. Tick the Remotable checkbox and save the object.
 - 5.4. Verify that your class is now remotable by clicking on Source card and checking that it has **@javax.xml.bind.annotation.XmlRootElement** annotation configured in the source code.

The screenshot shows the Business Central interface. The top bar includes 'Business Central', a home icon, a 'Menu' dropdown, and a user profile 'wbadmin'. The breadcrumb trail is 'Spaces > MySpace > myProject > master > PersonResponse'. The main area is titled 'PersonResponse.java - Data Objects' and contains tabs for 'Model', 'Overview', and 'Source'. The 'Model' tab is active, showing a table of fields for 'PersonResponse' with columns 'Identifier', 'Label', 'Type', and 'Delete'. The fields are 'name' (String) and 'age' (Integer). To the right of the table are 'general properties' for 'PersonResponse', including 'Identifier', 'Label', 'Description', 'Package' (com.myspace.myproject), and 'Superclass' (java.lang.Object). On the far right, the 'Drools & jBPM' panel is visible, containing various checkboxes and dropdowns. The 'Remotable' checkbox is checked and highlighted with a red box. The 'PersonResponse' label in the table is also highlighted with a red box. The 'Drools & jBPM' panel has a red box around the 'Drools & jBPM' header. At the bottom, an 'Alerts' table shows a successful build message for 'myProject'.

1. PersonResponse

2. Drools & jBPM

3. Remotable

Level	Text	File	Column	Line
0	Build of module 'myProject' (requested by system) completed. Build: SUCCESSFUL	-	0	0

